

Project Progress Report -Team 4

What Factors Affect Buldak Sauce Stain Removal Efficiency on White Cotton Fabric?

EXPERIMENTAL DESIGN AND ANALYSIS OF VARIANCE (001)

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1. Research Question & Objective

1.1 Research Question

This study originated from a focus on fabric staining issues common in everyday life. South Koreans frequently encounter red-pigmented dishes — kimchi, Malatang, and Buldak sauce — that easily discolor clothing and are notoriously difficult to launder. Identifying an effective removal method could prevent discarding garments or resorting to costly professional cleaning services.

Buldak sauce was selected as the target contaminant, motivated by its surging domestic and international popularity. Because it is a complex emulsion containing chicken extract, chili oil, soybean oil, and various spices, it presents a challenging and practically relevant stain type whose removal may depend on the chemical characteristics of the detergent, water temperature, and soaking duration.

Research question: Which combination of detergent type, water temperature, and soaking time most effectively removes Buldak sauce stains from white cotton fabric, as quantified by CIEDE2000 color difference (ΔE^*_{00})?

1.2 Experimental Objective

The objective is to evaluate the main effects and interaction effects of three factors on laundering efficiency:

- **Factor A:** Detergent type (dishwashing detergent vs. laundry detergent)
- **Factor B:** Water temperature (25°C vs. 60°C)
- **Factor C:** Soaking time (5 min vs. 30 min)

Response variable: ΔE^*_{00} (CIEDE2000 color difference between the stained fabric before and after laundering). A larger ΔE^*_{00} value indicates a greater color change, i.e., more effective stain removal.

2. Precise Definition of Treatments

Factor A: Detergent Type

- **Dishwashing detergent (D):** Saengong(생활공작소) brand dishwashing liquid. 5 g weighed on a digital scale and fully dissolved in 342.4 g of room-temperature tap water (25°C), yielding a 1.44% w/w solution.
- **Laundry detergent (L):** Saengong(생활공작소) brand liquid laundry detergent. 5 g dissolved in 342.4 g of room-temperature tap water (25°C) under the same protocol.

Rationale: The main components of Buldak sauce — chili oil and soybean oil — are lipid-based stains. Dishwashing detergents are specifically formulated to emulsify such lipid-based stains. Laundry detergents, on the other hand, are specialized for removing stains fixed to fabric, and we expect the comparison between the two to yield interesting results.

Factor B: Water Temperature

- **Cold (25°C):** Standard tap water. Temperature verified with a mercury thermometer immediately before immersion. Trials outside 23–27°C are discarded and repeated.
- **Hot (60°C):** Water boiled in an electric kettle and cooled to 60°C. Temperature measured immediately before immersion. Trials outside 58–62°C are discarded and repeated.

Rationale for level selection: The experimenter's washing machine offers 30°C, 40°C, 60°C, and 95°C settings. 95°C was excluded (fabric damage risk); 30°C was excluded (negligible difference from 25°C tap water); 40°C was excluded (cools to near-ambient too quickly during a 30-min soak). Thus 60°C was selected as the hot level.

Factor C: Soaking Time

- **Short (5 min):** Representing an immediate wash under time constraints. Soaking begins precisely 30 minutes after sauce application and continues for exactly 5 minutes, timed with a stopwatch.
- **Long (30 min):** Representing a deliberate pre-soak. Soaking begins at the same elapsed time and continues for 30 minutes.

Rationale: Soaking expands fabric fibers and softens fixed stain material, increasing laundering efficiency. The two levels contrast a time-constrained wash (5 min) with a thorough soaking protocol (30 min).

3. Precise Definition of the Response Variable

Response variable: ΔE^*_{00} (CIEDE2000 color difference)

CIEDE2000 is the current international standard for perceptually uniform color difference, incorporating corrections for lightness, chroma, and hue. Unlike simple mean RGB brightness or ΔL^* , it captures the full three-dimensional color change caused by laundering, including the reddish-orange hue of Buldak sauce. A smaller ΔE^*_{00} value indicates a smaller difference between the pre-contamination and post-wash states, corresponding to more effective stain removal.

Photography Protocol

- **Camera:** Smartphone in Pro mode with white balance and exposure locked throughout all trials.
- **Location:** Bathroom with door fully closed to block all ambient light.
- **Setup:** Paper foil on toilet lid, cutting board on top; experimenter's elbow rested on towel rack to fix camera position.
- **Distance:** Camera to cutting board fixed at 100 cm for all trials.
- **Timing:** Photos taken (i) immediately before stain application and (ii) after drying the following day, under identical conditions.

Calibration Procedure

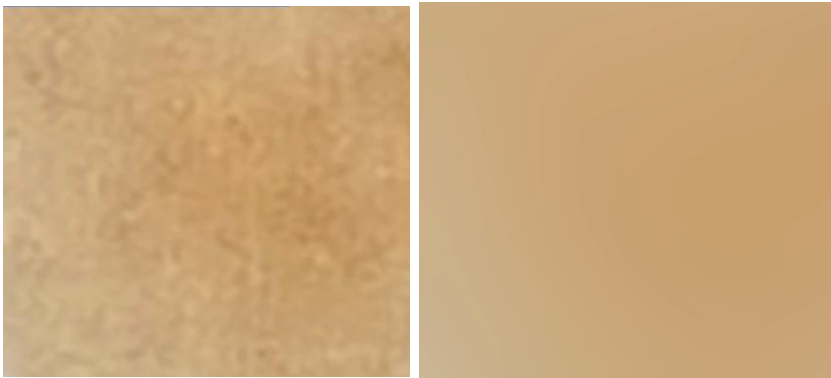
To correct for microscopic between-session lighting variation, a fixed calibration anchor point was established in every image: the center of the non-slip pad on the bottom-left corner of the cutting board. Both pre- and post-wash photos were digitally adjusted until the anchor-

point color matched its reference value.

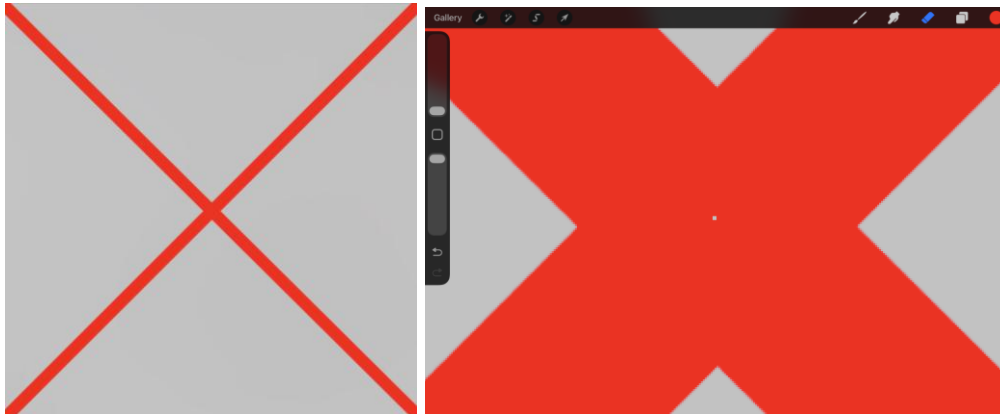
Calibration Point	RGB	CIE L*a*b*
Pre-contamination reference	(153, 152, 148)	L*=62.82 (a*=-0.39, b*=2.23)
Post-contamination reference	(150, 149, 147)	L*=61.73 (a*=-0.02, b*=1.18)

Color Extraction and ΔE^*_{00} Computation (R Code)

1. Apply Gaussian blur to the central region of the fabric in the calibrated image to smooth pixel noise.



2. Draw red guide lines connecting the four corners of the image; extract RGB from the central ROI.



3. Convert RGB to CIE L*a*b* in R using `farver::convert_colour(from="rgb", to="lab")`.
4. Compute CIEDE2000 per observation using `farver::compare_colour(from_space="lab", method="cie2000")`. This yields ΔE^*_{00} = color difference between clean and laundered states.

4. Experimental Design Details

Design Structure

A 2^3 full factorial design (3 two-level factors, all 8 treatment combinations) with 3 replicates per cell, yielding $N = 24$ runs in the pilot phase. The design will be expanded to 6 replicates ($N = 48$) in the main collection phase to achieve adequate power (see Section 6).

Factor	Low Level (-1)	High Level (+1)
A: Detergent Type	Dishwashing (D)	Laundry (L)
B: Water Temperature	25°C (cold)	60°C (hot)
C: Soaking Time	5 min	30 min

Experimental Units

24 pieces cut uniformly to 5 cm × 5 cm from the front face of the same white cotton T-shirt. The pieces are shuffled to destroy positional information, and numbered sequentially (1-1, 1-2, 2-1, ...) with permanent marker at a corner outside the ROI.

Randomization and Blocking

Twelve pairs of fabric pieces (n-1 and n-2) were randomly assigned to treatment combinations using Excel's =RAND() function to determine treatment assignment.

	A	B	C
1	0.58425	8	Dishwashing
2	0.89246	5	
3	0.137739	4	
4	0.950323	7	Laundry
5	0.800849	2	
6	0.812823	3	
7	0.684686	9	Dishwashing
8	0.020709	1	
9	0.314593	10	
10	0.375042	12	Laundry
11	0.570659	6	
12	0.389422	11	

Twin-set blocking: Pieces sharing the same index number (n-1 and n-2) form a twin set (Sets 1–12). Both pieces receive sauce simultaneously to control for staining variability. The n-1 piece is assigned to Container 1 (cold) and n-2 to Container 2 (hot).

Procedure

1. Cut the white cotton T-shirt into 24 pieces uniformly 5 cm × 5 cm.
2. Photograph all pieces in the standardized setting (clean baseline).
3. Start stopwatch; apply 5 g of Buldak sauce simultaneously to each twin set (n-1 and n-2); record application time.
4. Allow sauce to fix for exactly 30 minutes.
5. Measure and record temperatures of Container 1 (cold, 25°C) and Container 2 (hot, 60°C).
6. Per the random-number table: numbers 1–6 → 5-min soak; numbers 7–12 → 30-min soak.
7. Prepare detergent solution (5 g + 342.4 g water). Per the random-number table: numbers 1-3, 7-9 → Dishwashing detergent; numbers 4-6, 10-12 → Laundry detergent.

8. Add to container; agitate for 3 minutes at 80 BPM (metronome), switching hands at 1:30 to eliminate directional bias.
9. Rinse under running water for 3 minutes.
10. Dry all pieces in a single room overnight.
11. Photograph dried pieces the following day under identical conditions (laundered).

Nuisance Variable Control

- **Sauce fixation time:** Fixed at 30 minutes for all pieces.
- **Mechanical agitation:** 80 BPM metronome for 3 minutes; hands switched at 1:30 to remove directional bias.
- **Single experimenter:** All washing steps performed by one person to eliminate inter-experimenter variance.
- **Photography:** Fixed distance (100 cm), locked Pro-mode settings, same closed bathroom and lighting for all trials. Photographs were taken at the same time of day to ensure consistent ambient conditions.

5. Power Analysis

Inputs from pilot data (N = 24): Residual standard deviation $\sigma = 1.579 \Delta E^*_{00}$ units, taken from the MSE of the full factorial ANOVA model ($MSE = 2.493$, $df_2 = 16$). Power was computed in R using the non-central F distribution ($pf(\dots, ncp = \lambda, lower.tail = FALSE)$), with non-centrality parameter $\lambda = 2 \times n_{per_level} \times (\Delta/2)^2 / \sigma^2$.

Factor	Level Means (ΔE^*_{00})	Pilot Δ	Cohen's f	Power (N=24)	Power (N=48)
A: Detergent Type	D: 22.74 / L: 22.51	0.23	0.07	0.063	0.077
B: Water Temperature	25°C: 23.26 / 60°C: 21.99	1.27	0.40	0.456	0.775
C: Soaking Time	5 min: 22.71 / 30 min: 22.54	0.17	0.05	0.057	0.065
B×C Interaction (Temp×Time)	—	≈3.44*	—	p=0.017*	—

* The Temp×Time interaction was statistically significant in the pilot ANOVA ($F(1,16) = 7.11$, $p = 0.017$). The interaction contrast ($\Delta \approx 3.44 \Delta E^*_{00}$ units) represents the reversal of the temperature effect direction between 5-min and 30-min soaking conditions (see Section 9).

Achieving power ≥ 0.80 for the temperature main effect requires at least $r = 7$ replicates ($N = 56$; power = 0.838). The planned expansion to $r = 6$ ($N = 48$) yields power = 0.775, which falls slightly short of the conventional threshold but is accepted as a practical compromise given constraints on available materials and time.

Design revision: The current design ($N = 24$) is underpowered for all three main effects. The expansion to $N = 48$ is therefore planned, motivated primarily by replicating the significant Temp×Time interaction pattern rather than by main-effect power alone.

6. Ethics and Safety

This experiment was designed to be fully safe and to require no external ethical approval.

- **Participants:** No external human participants; only uniformly cut cotton fabric pieces are used as experimental units.
- **Contaminant and detergents:** Buldak sauce, dishwashing detergent, and laundry detergent are all commercially available household products with no chemical hazard at the quantities used.
- **Hot water:** The 60°C water condition is handled using thermally insulated gloves and a ladle to transfer water, preventing direct skin contact.
- No consent procedure, IRB review, or contingency plan is required for this experiment.

7. Analysis Plan

Formal Model

The response $\Delta E^*_{00,ijk\ell}$ (observation ℓ in cell ijk) is modeled by the full 2^3 factorial linear model:

$$\Delta E^*_{00} = \mu + \alpha_A + \alpha_B + \alpha_C + (\alpha\beta)_{AB} + (\alpha\beta)_{AC} + (\alpha\beta)_{BC} + (\alpha\beta\gamma)_{ABC} + \varepsilon$$

where $\varepsilon \sim N(0, \sigma^2)$ i.i.d. All three factors are fixed effects at two levels each.

Analysis Steps

1. Compute ΔE^*_{00} per run using `farver::compare_colour(..., method="cie2000")`.
2. Fit the additive model: `lm(deltaE ~ Type + Temp + Time, data=df)`; inspect ANOVA table.
3. Fit the full factorial model: `lm(deltaE ~ Type * Temp * Time, data=df)`; inspect ANOVA table for all main effects and interactions.
4. Construct a normal probability (Daniel) plot of the 7 effect estimates to visually identify dominant effects.
5. Produce interaction plots (`interaction.plot()`) and boxplots by factor level (`ggplot2::geom_boxplot()`) to visualize the Temp×Time interaction.
6. Residual diagnostics: residuals vs. fitted, normal Q-Q plot of residuals.

Software: R with packages `farver` (color conversion and ΔE^*_{00} computation) and `ggplot2` (visualization). ANOVA conducted with base `lm()` and `anova()`.

8. Pilot Data Summary and Preliminary Results

Response: ΔE^*_{00} (CIEDE2000), computed in R via `farver` from the raw RGB pixel values in the CSV. Larger values indicate more effective stain removal.

Soaking Time	Temp	Detergent	n	Mean ΔE^*_{00}	SD
5 min	25°C	Dishwashing (D)	3	24.585	2.903
5 min	25°C	Laundry (L)	3	23.819	0.458
5 min	60°C	Dishwashing (D)	3	21.388	1.341
5 min	60°C	Laundry (L)	3	21.040	1.659

30 min	25°C	Dishwashing (D)	3	22.501	1.753
30 min	25°C	Laundry (L)	3	22.125	0.789
30 min	60°C	Dishwashing (D)	3	22.472	1.747
30 min	60°C	Laundry (L)	3	23.056	0.109

Factor	Level	Marginal Mean ΔE^*_{00}
A: Detergent Type	Dishwashing (D)	22.74
A: Detergent Type	Laundry (L)	22.51
B: Temperature	25°C	23.26
B: Temperature	60°C	21.99
C: Soaking Time	5 min	22.71
C: Soaking Time	30 min	22.54

Pilot ANOVA Results

Additive model (main effects only):

Source	df	SS	MS	F	p-value
Type (Detergent)	1	0.307	0.307	0.104	0.750
Temp (Temperature)	1	9.653	9.653	3.266	0.086 .
Time (Soaking)	1	0.173	0.173	0.059	0.811
Residuals	20	59.109	2.955		

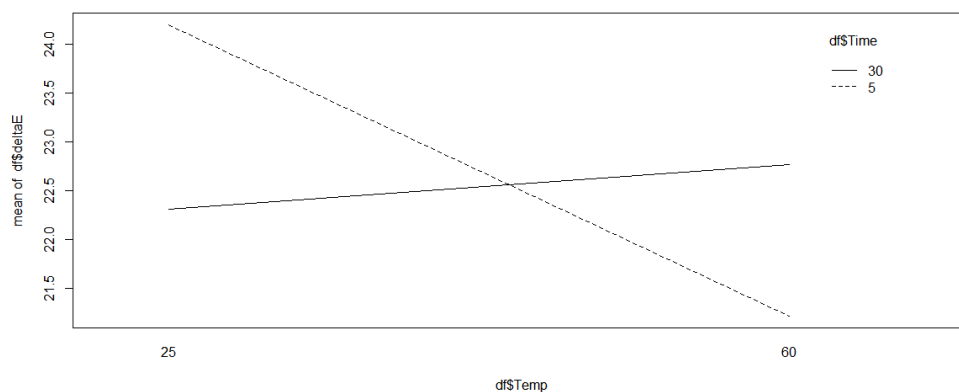
Full factorial model (all main effects and interactions):

Source	df	SS	MS	F	p-value
Type	1	0.307	0.307	0.123	0.730
Temp	1	9.653	9.653	3.871	0.067 .
Time	1	0.173	0.173	0.070	0.795
Type:Temp	1	0.712	0.712	0.285	0.601
Type:Time	1	0.654	0.654	0.262	0.615
Temp:Time	1	17.738	17.738	7.114	0.017 *
Type:Temp:Time	1	0.110	0.110	0.044	0.836
Residuals	16	39.895	2.493		

Key Finding: Temp × Time Interaction

The only statistically significant effect in the pilot is the Temp×Time interaction ($F(1,16) = 7.11, p = 0.017$). The interaction can be understood from the Temp×Time cell means (marginal over detergent type):

	25°C (Cold)	60°C (Hot)	Effect of temperature
5-min soak	24.20	21.21	-22.99 (hot → LESS removal)
30-min soak	22.31	22.76	+0.45 (hot → slightly more removal)
Effect of time	-1.89 (longer → less)	+1.55 (longer → more)	



At 5-minute soaking, hot water (60°C) produces substantially less stain removal than cold water (25°C). This counterintuitive result may reflect that hot water rapidly fixes the chili-oil pigment before the detergent can fully act in the short soaking window. At 30-minute soaking, the temperature effect reverses in direction and becomes negligible. This interaction pattern will be a central focus of the final analysis.

No main effects are statistically significant at $\alpha = 0.05$, although temperature shows a marginal trend ($p = 0.067$ in the full model). The absence of a significant main effect for soaking time ($\Delta = 0.17 \Delta E^*_{00}$ units) is consistent with the power analysis: the true main effect of soaking time may be very small, with the opposing interaction effects cancelling at the marginal level.

10. Modifications from the Proposal

- **Research question revision:** The original topic (seller's selling intention vs. buyer's purchase intention on secondhand platforms) was revised to the laundry experiment following IRB guidance regarding external human participants.
- **Single experimenter:** Initial design involved two experimenters; changed to a single experimenter to eliminate inter-experimenter variance in agitation.
- **Response variable:** Upgraded from mean RGB brightness to CIEDE2000 (ΔE^*_{00}) using R's farver package, to capture the full perceptual color change including hue and chroma shifts from the Buldak sauce pigment.
- **Agitation standardized:** A metronome at 80 BPM was introduced to standardize physical agitation; experimenter switches hands at 1:30 to remove directional bias.
- **Replicates to be increased:** Power analysis indicates $N = 24$ is underpowered. Additional 24 observations ($r = 4-6, N = 48$ total) will be collected by May 31, 2026.

11. Timeline for Remaining Work

Date	Milestone
May 26, 2026	Progress report submitted on Snowboard
May 31	Additional experiment: 24 new fabric pieces (replication sets 4–6) to reach $N = 48$. Each run: 30-min fixation + soaking + 3-min agitation + 3-min rinse + overnight drying.
June 1–2	Buffer days: re-run discarded trials (temperature out of range, photo failure)
June 3–5	Photograph dried post-wash pieces for sets 4–6; extract RGB, compute ΔE^*_{00} in R
June 6–8	Data entry, cleaning, and verification of all 48 ΔE^*_{00} values
June 9–12	Formal statistical analysis: ANOVA, Daniel plot, interaction plots, residual diagnostics
June 13	Final data analysis and figure generation
June 14	Statistical analysis write-up and infographic creation
June 15–18	Write Final Report (integrate progress report + results)
June 20	Complete final presentation slide deck and script
June 22	Final Project Presentation